Refactoring the GRC pipeline

What changed in RP7, and why

Emilia Tjernström

2026-05-04

The pipeline had drifted into twenty-two .do files we could not audit

- Twenty-two numbered .do files; experience and birth families spread across six of them
- A second Dropbox folder, `ReplicationPackage6 - real values/`, kept in sync with the nominal-values one by hand
- Ster filenames collided across .do files; some exceeded Stata's name limit

None of this changes the published numbers. All of it makes the pipeline harder to maintain.

We collapsed twenty-two .do files into thirteen

Before (RP6, 22 files)

- `5_GrRC.do`
- `6_GrRC_NonAg.do`
- `8_GrRC_hukou.do`
- `10_GrRC_experience.do`
- `11_GrRC_max_experience.do`
- `12_GrRC_experience_share.do`
- `13_GrRC_max_experience_share.do`
- `14_GrRC_NonAg_experience.do`
- `15_GrRC_birth.do`
- `16_heterogeneity_tables.do`
- ...

After (RP7, 13 files)

- `5_GrRC.do`
- `6_GrRC_NonAg.do`
- `8_GrRC_hukou.do`
- `GRC_extras.do` (experience family dispatcher)
- `make_tables.do`, `make_figures.do`
- `16_heterogeneity_tables.do`

Net: −9 files, same numbers

Experience and birth families: one program, one dispatcher

The five experience .do files plus `15_GrRC_birth.do` all did the same thing: GRC regression with one extra regressor.

What changed

- Shared regression logic: `run_grc_with_extra_regressor` in `0_programs.do`
- Per-cell calls: `GRC_extras.do`, one cell per (country, spec3, regressor)
- Tables: `extras_tex_table` reads sters and emits the family .tex files

To add a new family member

- One cell in `GRC_extras.do`
- One arm in `run_grc_with_extra_regressor`

One table-builder for all GRC tables

We had multiple table-building programs that were 95% the same code:

- `grc_tex_table_trend`
- `grc_tex_table_trend_exp`
- `grc_tex_table_trend_birth`
- `grc_tex_table_trend_hukou`

Now: one `grc_tex_table_trend`, four call signatures.

To change column widths, indicator-row order, postfoot string, or decimal precision, edit `grc_tex_table_trend` in `0_programs.do`.

One switch handles nominal and real values

Before

- Two parallel Dropbox folders
- Every edit propagated to both, by hand

Now

- One global, `values(nominal|real)`, in `0_path_config.do`
- `nominal`: filenames as before
- `real`: appends `_r` suffix; both modes coexist in one `output/` tree

To switch modes: edit `global values` in `0_path_config.do`.

How ster filenames are built

Pattern: `grc_<country>_<spec3>_<covs2>[_<sfx1>].ster`

`country`	`IDN`, `CHN`, `TZA`, or `CHN_hukou_<subgroup>`
`spec3`	`cuu`, `cub`, `iuu`, `cnu`
`covs2`	`c0`, `ct`, `c1`, `c2`, `ca`
`sfx1`	`n` (never), `a` (always), `d` (delta), `g` (average)

Every ster name is built from these tokens. Look up any fit by combining them.

Full convention in `RP7/scripts/STER_NAMING.md`.

Editing the GRC tables

To change caption, label, or notes

- Edit the `\GRCtable{}` and `\GRChukoutable{}` macros in `preamble.tex` (the paper)

To change which coefficients appear, label text, or formatting

- Per-cell options: `make_tables.do`
- Underlying template: `grc_tex_table_trend` in `0_programs.do`

Cosmetic refresh

- Average Δ row reads $\bar{\Delta}$
- Empty `esttab` rows now stripped

What this buys us

- One source of truth for the regression code
- One source of truth for the table caption, label, and notes (in the paper preamble)
- Ster filenames you can read, type, and look up
- Same published numbers as before

The pipeline is now small enough to hold in one head.